



PARTS LIST

Semiconductors:

IC1	- LM7812
IC2 (IC2A-IC2C)	- LM358 op-amp
T1	- BC547 npn transistor
BR1	- DB107, 1A bridge rectifier
D1	- 1N4007 rectifier diode
D2	- 1N4148 signal diode
ZD1	- 2.2V zener diode
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 47-kilo-ohm
R2	- 2.2-mega-ohm
R3, R4	- 100-kilo-ohm
R5, R6	- 10-kilo-ohm
R7	- 1-kilo-ohm
R8	- 2.2-kilo-ohm
VR1	- 100-kilo-ohm potmeter

Capacitors:

C1, C2, C4	- 220 μ F, 25V electrolytic
C3	- 4700 μ F, 25V electrolytic

Miscellaneous:

X1	- 230V AC primary to 12V, 500mA secondary transformer
RL1	- 12V, 10A SPDT relay
S1	- Push-to-on switch
S2	- 10A SPDT switch
CON1	- 6A, 230V AC cord
CON2	- 6A, 230V plug socket
Enclosure	- Suitable-size PVC enclosure
PCB board	- 10.2cm x 7.6cm (4"x3") veroboard

with current sourcing and sinking capability. The positive terminal voltage of capacitor C3 is fed to pin 3 via resistor R3, and so the same voltage appears at pin 1. 2.2V zener diode is connected between pin 1 and point J (the junction of R1 and R2), and thus the voltage at J remains 2.2V above the voltage at pin 1.

Capacitor C3 charges slowly through resistors R1 and R2. Initially, the capacitor voltage is zero,

making the voltage at pin 1 also zero and that at 'J' 2.2V. As C3 charges, its voltage, as well as the voltage at pin 1 and J, increases. If the capacitor voltage increases by ΔV , the voltages at pin 3, pin 1, and J also increase by the same voltage ΔV , and the voltage at J is $2.2 + \Delta V$. Thus, the charging current through R2 is:

$$\begin{aligned} & \{ (2.2 + \Delta V) - \Delta V \} \div R2 \\ &= 2.2 \div (2.2 \times 10^6) \text{ ampere} \\ &= 0.001 \text{mA} \end{aligned}$$

This current remains fairly constant, ensuring a linear voltage rise in the capacitor. The voltage of capacitor C3 is fed to the non-inverting input pin 5 of op-amp IC2B through resistor R4. The inverting input pin 6 voltage is set by potentiometer VR1, which acts as the time control. When the voltage at pin 5 exceeds the voltage at pin 6, the output pin 7 goes high, making transistor BC547 (T1) conduct and energise relay RL1. The higher the voltage setting of VR1, the longer time is required for the changeover.

Two LEDs, LED1 and LED2, show the status of power and timer operation. The capacitor C4 and diode D2 filter out any possible relay chattering during the transition of the op-amp as the voltage rise in C3 is very slow.

Construction and testing

The circuit can be assembled on a 10.2cm x 7.6cm (4"x3") veroboard and housed in a suitable-size PVC

enclosure with a power cord and plug socket. The appliance to be controlled is connected to this plug socket, for receiving or stopping power at the set time.

SPDT switch S2 acts as a selector switch for selecting on/off timer. Switch S1 starts or resets the timer. Momentarily pressing S1 discharges C3, which then begins charging from zero.

Potentiometer VR1 can be calibrated in a linear time scale from 0 to 12 hours. The maximum time can be changed by altering the values of C3, R2, and zener diode. Diode D1 acts as the freewheeling diode to bypass the inductive spike of the relay coil.

Note. Transformer X1, rectifier BR1, capacitors C1 and C2 and IC1 can be replaced by an SMPS-based 12V power supply, which will reduce the component count as well as the size of the PCB.

2. Two 1N4007 diodes in series (so that current flows from J to Pin 1) may be used in place of the zener diode ZD1 to charge C3 at 1.4V, which will increase the timer's duration to more than 24 hours. **EFY**

Pankaj Datta Choudhury is DGM (Tech), Tripura Gas Based Power Station (TGBPS), North Eastern Electric Power Corporation Ltd (NEEPCO), Monarchak, Tripura. A gas turbine ignition exciter pulse generator designed by him is in use at TGBPS and he has applied for patent for a smart illumination controller designed by him.



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